



Multiple exponence in child language

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Introduction

Children produce overregularization errors during acquisition (Marcus et al. 1992), which can result in **redundant errors**

(1) English past tense

ate > eat-ed > **ate-d** > ate (Kuczaj 1977, 1978)

What is the nature of redundant errors like *ate-d*?

- **Multiple exponence:** A single feature PAST is incorrectly realized twice, once by the stem *ate-* and once by the affix *-ed*
- **Allomorphy:** *ate* is learned as an allomorph of EAT in the context of PAST, which is incorrectly spelled out as *-ed*
- **Radical decomposition:** *ate* spells out EAT and some other (non-PAST) feature, PAST is incorrectly spelled out as *-ed*

Commission error: Misapplication of a rule or generalization (see e.g. Snyder 2007, 2011)

- **Distributive** commission error: *eat-ed* (eat-PAST)
- **Redundant** commission error: *ate-d* (eat.PAST-PAST)

Claim 1

Redundant commission errors in child language involve true multiple exponence.

Claim 2

Multiple exponence can occur **across word boundaries**, thus favouring models of morphology that do not grant independent ontological status to the concept of word.

1. Emergence of commission errors
2. Case studies
 - Causative
 - Comparative
3. Theoretical analyses
 - Nanosyntax: Spanning
 - Distributed Morphology: Fusion
4. Extension
 - Locative prepositions

Emergence of commission errors

Commission errors

Commission errors across domains (see also Martin et al. 2021)

Commission errors

Target form	Distributive error	Redundant error
<i>ate</i>	<i>eat-ed</i> EAT PAST	<i>ate-d</i> EAT.PAST PAST
<i>mieux</i> 'better'	<i>plus bon</i> COMP GOOD	<i>plus mieux</i> COMP COMP.GOOD
<i>fermer</i> 'to close'	–	<i>faire fermer</i> CAUSE CAUSE.CLOSED
<i>sous</i> 'under'	–	<i>dessous</i> PLACE PLACE.AXPART.EIGEN
<i>rien</i> 'nothing'	<i>pas...quelque chose</i> NEG...∃	<i>pas...rien</i> NEG NEG.∃

Principles of exponence

Adult grammar

Minimize Exponence! (Siddiqi 2006)

Realize a set of concepts using the fewest exponents.

- General economy principle, e.g. EAT, PAST → *ate*

Child grammar

- **One-to-one mapping principle** (Slobin 1973, van Hout 2008, Alexiadou et al. 2021)
- General transparency principle, e.g. EAT, PAST → *eat-ed*

Maximize Exponence!

Realize each concept using exactly one exponent.

Principles of exponence

Language acquisition

- Two competing principles

Maximize Exponence!

Realize each concept using exactly one exponent.



Multiple Exponence



Minimize Exponence!

Realize concepts using the fewest exponents.

- Intermediate stage of **Multiple Exponence** where transparent, decomposed forms exist alongside non-transparent, portmanteau forms
 - e.g. EAT, PAST → *ate-d*

Predictions for development

- Children first decompose in a one-to-one fashion (**stage 1**)
→ Distributive commission errors, e.g. *eat-ed*
- This pressure for transparency perseveres even after they have started acquiring target portmanteau forms (**stage 2**)
→ Redundant commission errors, e.g. *ate-d*
- Multiple exponence in child grammar reflects the attempt to maximize both transparency and economy
- e.g. EAT, PAST → *eat-ed* > *ate-d* > *ate*
COMP, GOOD → *plus bon* > *plus mieux* > *mieux*
- Multiple exponence may persist in non-standard adult language, either as uncontrolled redundant errors, or as controlled ways to emphasize some aspect of meaning

Commission errors

Stages of exponence

Distributive error <i>Maximize Exponence!</i>	Redundant error Multiple Exponence	Target form <i>Minimize Exponence!</i>
<i>eat-ed</i> EAT PAST	<i>ate-d</i> EAT.PAST PAST	<i>ate</i>
<i>plus bon</i> COMP GOOD	<i>plus mieux</i> COMP COMP.GOOD	<i>mieux</i> 'better'
–	<i>faire fermer</i> CAUSE CAUSE.CLOSED	<i>fermer</i> 'close'
–	<i>de</i> <i>essous</i> PLACE PLACE.AXPART.EIGEN	<i>sous</i> 'under'
<i>pas...quelque chose</i> NEG...∃	<i>pas...rien</i> NEG NEG.∃	<i>rien</i> 'nothing'

Case studies

- Two case studies of redundant commission errors in child French and English
 1. Causative: *faire fermer* 'CAUSE CAUSE.CLOSED'
 2. Comparative: *plus mieux* 'COMP COMP.GOOD'
- These cases are instances of true multiple exponence, which furthermore occur across word boundaries

- Lexical causative verbs encode a causative meaning component CAUSE, e.g. French *fermer* 'close' or *montrer* 'show'
- Productive causatives can be formed in French using the verb *faire* 'make', which encodes an additional CAUSE component

(2) a. *Montre le camion de pompiers.*

'Show the firetruck.'

b. *J'ai fait montrer le camion au client par un de nos meilleurs vendeurs.*

'I made one of our best salesmen show the truck to the client.'

- Children in French CHILDES corpora (MacWhinney 2000) frequently produce lexical causatives with a redundant *faire*, thus spelling out CAUSE twice (Martin et al. 2021)

(3) a. *faire fermer les yeux.* (LSN 4;02, Palasis 2009)

Intended meaning: 'Close the eyes.'

b. *va le faire couper.*

Intended meaning: 'Going to cut it.'

(Marilyn 2;09, Demuth and Tremblay 2008)

c. *du bon feu ici pour les faire réchauffer.*

Intended meaning: 'A nice fire here for reheating them.'

(Camille 3;09, Le Normand 1986)

Causatives

- Matteo (Palasis 2009) and Madeleine (Morgenstern et al. 2009) use the portmanteau lexical causative form before/alongside the redundant form
- Provides evidence for a multiple exponence analysis

- (4) a. *Elle a **fait tomber** ma petite cabane.* (Matteo 2;11)
'She made my little shed fall.'
- b. *J'ai **montré** ça.* 'I showed that.' (Matteo 3;02)
- c. *Eh **fais montrer** le camion de pompiers!* (Matteo 3;03)
Intended meaning: 'Hey show the firetruck!'
- (5) (a) *près on va le **cacher** ... on va le **cacher** ... va le **faire cacher**.* (Madeleine 2;02)
Lit.: 'Then we'll hide it ... we'll hide it ... we'll make hide it.'

- Multiple exponence of CAUSE is attested in
 - child French (Bezinska et al. 2008, Martin et al. 2021)
 - child Turkish (Aksu-Koç and Slobin 1985)
 - child Persian (Family and Allen 2015)
 - child Japanese (Yamakoshi et al. 2018)
 - child English (Lord 1979, Nie et al. in progress)
- It also arises in non-standard and dialectal adult variants

Examples of CAUSE multiply realized in non-standard adult French:

- (6) a. *Ah mais tu m'a [sic] fait donner une idée.*
'Oh but you gave me an idea.' (Twitter)
- b. *une blessure va lui faire montrer un tout autre chemin*
'An injury will show her a totally different path.'
(corpus frTenTen17 via Sketchengine)
- c. *J'ai fais [sic] fermer mes yeux il était 1h59 mtn il est 3h00 mdr*
'I closed my eyes it was 1.59 now it's 3.00 lol.'
(Twitter)

Comparatives

- Regular comparatives in adult French are formed with word *plus* 'more' and the positive form of the adjective
- For *bon* 'good', there is a portmanteau comparative form *mieux* 'better' which blocks the regular form **plus bon*
- French children frequently produce *mieux* with a redundant *plus* (cf. Moline 1971), thus spelling out COMP twice

- (7) a. C'est *plus mieux* comme ça. 'It's more better like this.'
- b. on va i donner un petit peu d'eau (...) pour qu'i soit *plus mieux*.
'We'll give him a little bit of water (...) so that he's feels more better.' (VET, Saint-Pierre and Feider 1987)

Comparatives

- Suppletive and regular -er suffixed comparative forms can appear with redundant *more* (or -er) in child English

(8) a. *I like toasts more better.* (Abe 4;03, Kuczaj 1977)

b. *I make it more bigger.*

(Roman 3;09, Weist and Zevenbergen 2008)

c. *that's even lighterer.* (Helen 4;02, Lieven et al. 2009)

(9) *MAR: *a little bit more drier .*

*FAT: *yeah (.) that's true (.) and cleaner (.) right .*

*MAR: *and more drier and more cleaner .*

(Mark 3;09, MacWhinney 2000)

- Children produce the target comparative form before/alongside the redundant form
- Redundant comparatives can be overtly marked twice in English, with *more* and *-er*
- No correlation between irregular formation of the comparative and likelihood of commissive *more* in child English
 - e.g. *better* is no more likely than *bigger* to occur with commissive *more*
- Provides evidence for a multiple exponence analysis

Generalizations

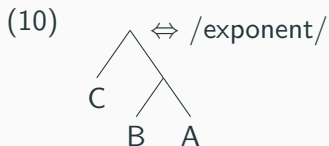
- Redundant commission errors involve multiple exponence and can occur across word boundaries
- The element that is multiply realized tends to be
 - A higher element in the projection
 - A functional element, rather than the root
- Multiple exponence of lower elements are unattested or rare and do not seem to persist in the adult grammar

Redundant commission error		
Target form	well attested	unattested/rare
<i>donner</i>	<i>faire donner</i>	* <i>donner avoir</i>
CAUSE HAVE	CAUSE CAUSE.HAVE	CAUSE.HAVE HAVE
<i>mieux</i>	<i>plus mieux</i>	* <i>mieux bon</i>
COMP GOOD	COMP COMP.GOOD	COMP.GOOD GOOD

Theoretical analyses

- Data show that ME occurs across word boundaries.
- Challenge for approaches that separate word-formation from phrase-formation (e.g. Paradigm Function Morphology, Stump 2001, 2016)
- Syntax-based approaches to morphology should be able to account for the data
- Two analyses presented here:
 - Nanosyntax (Starke 2009, Caha 2009, et seq.)
 - Distributed Morphology (Halle and Marantz 1993, 1994)

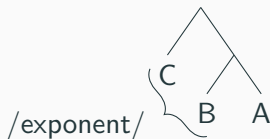
- Nanosyntax (Starke 2009, Caha 2009, et seq.) allows non-terminal spellout, i.e. spellout out of several terminal nodes that form a constituent at once.



- Lexicalization follows the Superset Principle. Previous lexicalizations may be overridden by subsequent lexicalizations.

- Spanning (Williams 2003, Abels and Muriungi 2008, Taraldsen 2010, Svenonius 2012, a.o.) allows lexical items to spell out non-constituents (span = “a contiguous sequence of heads in a head-complement relation”, Svenonius 2016: 205).

(11)



Claim

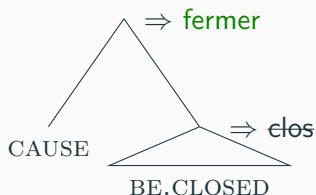
Children's commission errors result from **erroneous overlapping application** of spanning lexicalization (S-lexicalization) and run-of-the-mill constituent lexicalization (C-lexicalization).

(12) Lexical items

a. $\text{clos} \Leftrightarrow [\text{BE.CLOSED}]$

b. $\text{former} \Leftrightarrow [\text{CAUSE} [\text{BE.CLOSED}]]$

(13) C-lexicalization overrides previous C-lexicalization



Nanosyntax: Commissive faire fermer I

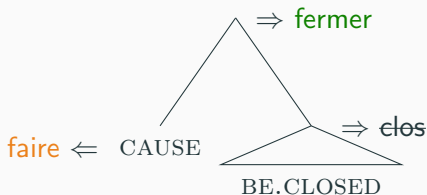
(14) Lexical items

a. *clos* $\Leftrightarrow$ [BE.CLOSED]

b. *fermer* $\Leftrightarrow$ [CAUSE [BE.CLOSED]]

c. *faire* $\Leftrightarrow$ [CAUSE

(15) Overriding *clos* with *fermer* and (erroneous) S-lexicalization of CAUSE by *faire*



Nanosyntax: Commissive faire fermer II

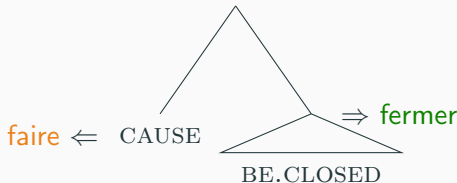
(16) Lexical items

a. clos $\Leftrightarrow$ [BE.CLOSED]

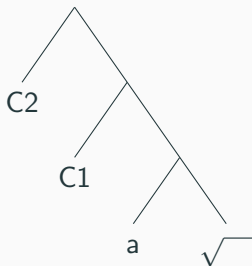
b. fermer $\Leftrightarrow$ [CAUSE [BE.CLOSED]]

c. faire $\Leftrightarrow$ [CAUSE

(17) Elsewhere error inserting *fermer*, failure to override *fermer* and S-lexicalization of CAUSE with *faire*



(18) Structure of comparative phrase (Caha et al. 2019)

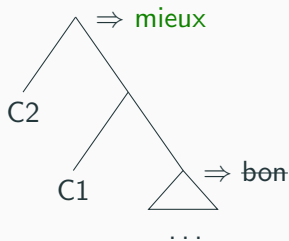


(19) Lexical items

a. $\text{bon} \Leftrightarrow [\text{a } \sqrt{\text{GOOD}}]$

b. $\text{mieux} \Leftrightarrow [\text{C2 } [\text{C1 } [\text{a } \sqrt{\text{GOOD}}]]]$

(20) Overriding *bon* with *mieux*



Nanosyntax: Commissive 'plus mieux'

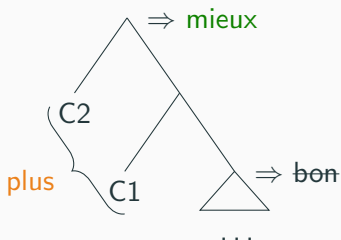
(21) Lexical items

a. $\text{bon} \Leftrightarrow [\text{a } \sqrt{\text{GOOD}}]$

b. $\text{mieux} \Leftrightarrow [\text{C2 } [\text{C1 } [\text{a } \sqrt{\text{GOOD}}]]]$

c. $\text{plus} \Leftrightarrow [\text{C2 } [\text{C1}$

(22) Overriding *bon* with *mieux* and (erroneous) S-lexicalization of $[\text{C1}[\text{C2}$ by *plus*



Nanosyntax: Commissive 'more bigger'

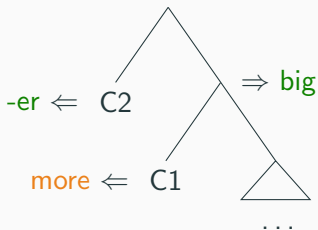
(23) Lexical items

a. $\text{big} \Leftrightarrow [\text{C1 } [\text{a } \sqrt{\text{BIG}}]]$

b. $\text{more} \Leftrightarrow [\text{C2 } [\text{C1}$

c. $\text{-er} \Leftrightarrow [\text{C2}$

(24) (Erroneous) S-lexicalization of C1 by *more*



- Vocabulary items may only be inserted into terminal nodes (heads) following the Subset Principle and Specificity.
- Head movement may generate complex heads, whose terminals may further undergo Fusion resulting in all features involved being present on a single simplex head.
- The application of Fusion is regulated by Minimize Exponence, a (transderivational) constraint that prefers the derivation that uses fewer exponents over one that uses more exponents to realize the same features (Siddiqi 2006).

Claim

Vocabulary Insertion takes place in accordance with the Specificity Principle but may **fail to discharge some features** (Maximize Exponence) which may then affect subsequent insertions.

DM: Causative 'fermer'

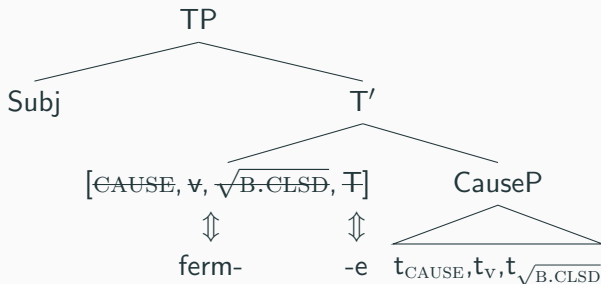
(25) Vocabulary items

a. *fermer* $\leftrightarrow$ $[\text{CAUSE}, \mathbf{v}, \sqrt{\text{BE.CLOSED}}]$

b. *clos* $\leftrightarrow$ $[\mathbf{v}, \sqrt{\text{BE.CLOSED}}]$

c. *faire* $\leftrightarrow$ $[\text{CAUSE}]$

(26)



DM: Commissive 'faire fermer'

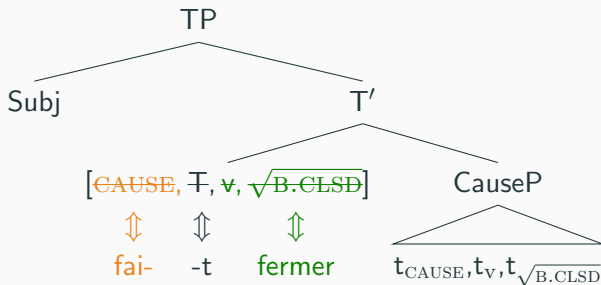
(27) Vocabulary items

a. *fermer* $\leftrightarrow$ [CAUSE, v, $\sqrt{\text{BE.CLOSED}}$]

b. *clos* $\leftrightarrow$ [v, $\sqrt{\text{BE.CLOSED}}$]

c. *faire* $\leftrightarrow$ [CAUSE]

(28)

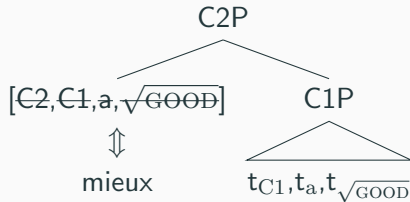


(29) *Vocabulary items*

a. *mieux* $\leftrightarrow [C2, C1, a, \sqrt{\text{GOOD}}]$

b. *bon* $\leftrightarrow [a, \sqrt{\text{GOOD}}]$

(30)



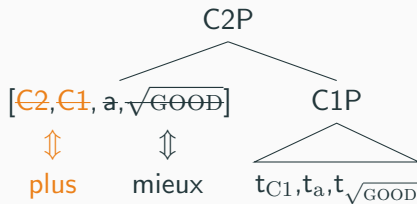
(31) *Vocabulary items*

a. *mieux* $\leftrightarrow$ $[C2, C1, a, \sqrt{\text{GOOD}}]$

b. *bon* $\leftrightarrow$ $[a, \sqrt{\text{GOOD}}]$

c. *plus* $\leftrightarrow$ $[C2, C1]$

(32)



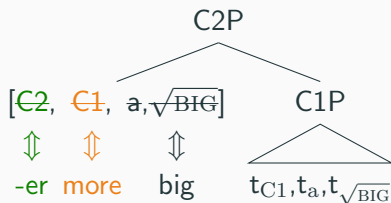
(33) *Vocabulary items*

a. *big* $\leftrightarrow$ $[C1, a, \sqrt{\text{BIG}}]$

b. *-er* $\leftrightarrow$ $[C2]$

c. *more* $\leftrightarrow$ $[C2, C1]$

(34)



DM:

- Without further restrictions *mieux* could equally well discharge only [C2,C1]. We would expect [a, $\sqrt{\text{GOOD}}$] to be realized by *bon* resulting in unattested **mieux bon*.
- We cannot appeal to directionality (bottom-up) as all features are fused on one terminal without hierarchical order.

Nanosyntax:

- In overlapping C-lexicalization and S-lexicalization, the latter applies exclusively to higher material.
- Correctly predicts multiple exponence of high/functional elements rather than root material.
- Linearization doesn't seem to be straightforward.

Extension

- Some cases of multiple exponence in child language only involve functional material
- It is again the higher element which tends to be doubled
- Locative prepositions

Locative prepositions

- Svenonius (2006): Locative prepositions have a structure consisting of three locative concepts
 - English *behind, in, under, above*
 - French *derrière, dans, sous, sur*

(35) a. [PLACE [AXPART [EIGEN [GROUND]]]]

b. be -hind Ø the car

c. de -rrière Ø la voiture

- Some child languages exhibit multiple exponence of the PLACE concept

Locative prepositions

- French *sous* 'under' spells out the three locative concepts PLACE, AXPART, EIGEN
- *sous* can occur with redundant *de* in child French, thus spelling out PLACE twice
 - *dessous* exists as an independent preposition in French but cannot be used with a following DP

(36) *MOT: i(l)s sont cachés **sous** les ailes.
 they are hidden under the wings

(37) *CHI: est caché **dessous** les ailes.
 is hidden PLACE.under the wings

‘They are hidden under the wings.’

(Théotime 2;05, Demuth and Tremblay 2008)

Conclusion

- Redundant commission errors are pervasive in child language
- These errors can occur across word boundaries and should be considered true cases of multiple exponence
- Cross-word multiple exponence favors syntactic approaches to word formation
 - Multiple exponence of high/functional elements is straightforwardly predicted in a Nanosyntactic approach
- Multiple exponence in child language reflects an intermediate stage of acquisition in which children attempt to maximize both transparency and economy



Z A S



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